

Oil Spill Detection and Vessel Attribution

Goal: Detect suspicious marine oil slicks, characterize their geometry, estimate a probable origin region and time window, then rank vessels using spatio-temporal AIS evidence.

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Problem Statement 26143

Review 1

01 Oil spill: the problem

What is an oil spill?

An oil spill is the accidental or deliberate release of crude oil, fuel or petroleum products into the ocean or other water bodies.

Why does it occur?

- ▶ Tanker and ship accidents.
- ▶ Pipeline or offshore platform failures.
- ▶ Illegal discharge from vessels.
- ▶ Drilling and loading operations.
- ▶ Equipment failure and human error.

Impact on marine life

- ▶ Oil coats feathers and fur, reducing insulation and mobility.
- ▶ Toxic compounds can poison fish, birds and marine organisms.
- ▶ Oil can damage coral reefs, mangroves and coastal habitats.
- ▶ Contaminated food chains can disrupt marine ecosystems.
- ▶ Long-term contamination can affect reproduction and survival.

Our objective

Detect the spill early and identify likely sources using evidence.

02 Existing approaches and the research gap

Traditional

- ▶ Visual interpretation.
- ▶ Intensity or threshold rules.
- ▶ Manual polygon extraction.

Weakness

- ▶ Sensitive to imaging conditions.
- ▶ Difficult to scale.
- ▶ High false positives.

Deep learning

- ▶ Learns spatial context.
- ▶ Uses texture, shape and polarization.
- ▶ Produces pixel-level masks.

Weakness

- ▶ Needs representative labels.
- ▶ Can suffer from domain shift.

Operational attribution

- ▶ AIS gives vessel trajectories.
- ▶ Drift models constrain possible origins.
- ▶ Combining both gives physical evidence.

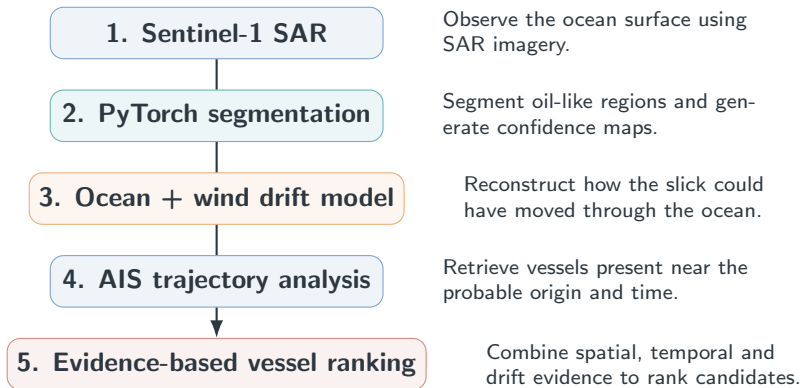
Weakness

- ▶ Detection alone does not identify a source.
- ▶ Nearest-vessel matching ignores drift and time.

Core gap

Connect image evidence to physical origin and vessel history.

A five-stage evidence pipeline



04 What exactly do we detect?

OIL SLICK

- ▶ Oil-like SAR signature.
- ▶ Pixel mask.
- ▶ Area and centroid.
- ▶ Perimeter and orientation.
- ▶ Elongation.

LOOK-ALIKE

- ▶ Low-wind areas.
- ▶ Biogenic films.
- ▶ Rain-related effects.
- ▶ Internal waves.
- ▶ Vessel wakes.

NO OIL

- ▶ Normal ocean background.
- ▶ Prevents “dark = oil” learning.
- ▶ Supports false-positive control.

[SAR patch → Oil probability + Segmentation mask + Confidence]

Why DL? The model learns context, texture, shape and multi-channel patterns that a single intensity threshold cannot represent.

05 Three data streams

Data	What it gives	Acquisition	Role
Sentinel-1 SAR	VV/VH backscatter, metadata, oil masks	Public Sentinel-1 products and research datasets	Detection + geometry
AIS	MMSI, timestamp, latitude, longitude, SOG, COG, vessel metadata	MarineCadastre or another permitted AIS source	Vessel history
Ocean + wind	Surface current vectors and wind forcing	Copernicus Marine + ERA5 or equivalent	Drift simulation

06 Dataset structure and usage

Sentinel-1 oil-spill benchmark

- ▶ Oil-spill images with corresponding segmentation masks.
- ▶ VV/VH SAR channels where available.
- ▶ Oil, no-oil and look-alike examples.
- ▶ Held-out test material for evaluation.

Usage

[Train → Validate → Test]

AIS dataset

- ▶ Vessel identity through MMSI.
- ▶ Time-stamped position history.
- ▶ Speed and course.
- ▶ Vessel metadata where available.

Environmental data

- ▶ Current vector field.
- ▶ Wind vector field.
- ▶ Interpolated to simulation time and location.

07 Preprocessing and model training

SAR preprocessing

1. Validate scene metadata.
2. Select VV/VH channels.
3. Normalize consistently.
4. Tile large scenes.
5. Preserve image-mask alignment.
6. Augment training data only.

PREVENT LEAKAGE

PRESERVE ALIGNMENT

PROTECT TEST SET

Training setup

1. PyTorch Dataset and DataLoader.
2. Train / validation / test separation.
3. U-Net baseline for segmentation.
4. BCE + Dice loss.
5. Save best validation checkpoint.
6. Evaluate on held-out test data.

Compute strategy: Tile large scenes, train on small batches, use transfer learning where appropriate, and process the full archive only during inference.

08 Deep-learning model: PyTorch U-Net baseline

Input

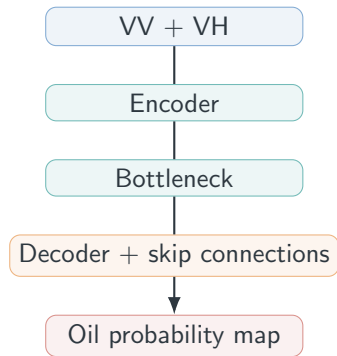
- ▶ SAR patch, typically VV + VH.
- ▶ Fixed-size training tiles.

Network

- ▶ Encoder extracts hierarchical features.
- ▶ Bottleneck captures high-level context.
- ▶ Decoder restores spatial resolution.
- ▶ Skip connections preserve slick boundaries.

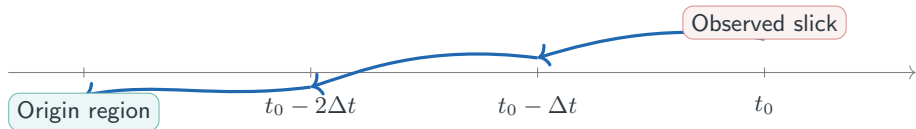
Output

- ▶ Pixel-level oil probability map.
- ▶ Thresholded segmentation mask.



09 Backtracking: estimating the probable origin

Core idea: Start from the observed slick and integrate representative particles backward through the environmental field.



- ▶ Use a Lagrangian particle model, such as OpenDrift, or a validated custom implementation.
- ▶ Run backward trajectories to estimate an origin probability region.
- ▶ Run forward trajectories to support short-term slick forecasting.

Output: A probable origin distribution, not an exact historical point.

10 AIS correlation and vessel ranking

1. Spatial filter

- ▶ Find vessels near the origin uncertainty region.
- ▶ Keep a configurable search radius.

2. Temporal filter

- ▶ Compare vessel presence with the estimated release window.
- ▶ Reconstruct vessel trajectory around that window.

3. Physical consistency

- ▶ Check whether the observed slick could drift from the candidate location to the observed location.

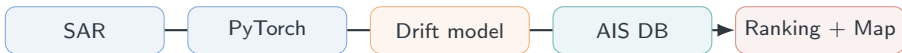
Evidence score

$$[S_v = w_s S_{space} + w_t S_{time} + w_d S_{drift} + w_b S_{behavior}]$$

- ▶ Spatial proximity to origin.
- ▶ Temporal compatibility.
- ▶ Drift consistency.
- ▶ Optional behavior anomalies.

Rank candidates, do not accuse automatically.

11 System architecture and technology stack



ML / CV

- ▶ Python
- ▶ PyTorch
- ▶ NumPy
- ▶ OpenCV
- ▶ scikit-learn

Geospatial

- ▶ Rasterio
- ▶ GeoPandas
- ▶ Shapely
- ▶ xarray
- ▶ OpenDrift

Backend / UI

- ▶ FastAPI
- ▶ PostgreSQL + PostGIS
- ▶ React
- ▶ Leaflet / MapLibre
- ▶ Docker

Prototype principle: Each module produces inspectable outputs for the next module.

12 Novelty, validation and feasibility

Novelty

- ▶ Explicit oil versus look-alike learning.
- ▶ Physics-informed origin estimation.
- ▶ Backward + forward slick reasoning.
- ▶ Spatio-temporal AIS candidate scoring.
- ▶ Evidence display instead of black-box accusation.

Validation

- ▶ Segmentation: IoU, Dice, precision, recall.
- ▶ Origin: localization error in km.
- ▶ Time: origin-time error.
- ▶ Attribution: Top-1 / Top-k recovery on controlled cases.

Feasibility

- ▶ Public research data can support the prototype.
- ▶ Open-source ML and geospatial tools.
- ▶ Tile-based training reduces compute requirements.
- ▶ Modular architecture supports incremental development.

NOVEL

MEASURABLE

FEASIBLE